

**UNIVERSITY OF ILLINOIS-CHICAGO
DEPARTMENT OF CHEMICAL ENGINEERING**

**CHE 396 – SENIOR DESIGN I
Fall 2025**

Lecture:

Section 1: M/W: 11:00am – 12:15pm: in Room EIB 124

No Classes:

September 1st (Labor Day)

Office Hours:

Mondays/Wednesdays 12.15pm-1.15pm in room EIB 144

Instructor:

Dr. Betul Bilgin, bbilgin@uic.edu

Teach Asst:

None

Prerequisites:

CHE 312, CHE 313, and CHE 312

Text:

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012. (Textbook website: booksite.elsevier.com/Towler) (required)

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009 (as a reference)

Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker (printed book ISBN: 9781932780093 / retail price: \$39 or e-book ISBN: 9781932780109 / a retail price \$19) (as a reference)

Course Description:

This is an introduction class to modern process design and development, engineering economics, and report writing. It also covers the design and cost of equipment related to materials handling to heat transfer and reactors.

Course Objectives:

- Provide a real-world perspective on process design

- Introduce design and selection of equipment
- Give insight into practical constraints that govern design decisions
- Help you translate knowledge into practice & start solving real problems
- Improve presentation, analytical thinking, and team working skills

Course Outline:

Date	Chapter	Assignment
August 25 th 2025	Syllabus Development	
August 27 th 2025	Chapter 1 (Introduction)	
September 1 st 2025	NO Class	
September 3 rd 2025	Chapter 2 (Flowsheet development)	
September 8 th 2025	Chapter 2 (Flowsheet development)	
September 10 th 2025	Librarian Guest Lecture	
September 15 th 2025	Chapter 2 (Flowsheet development)	1 st Assignment Due (September 15 th)
September 17 th 2025	Presentation 1	
September 22 nd 2025	Presentation 1	
September 24 th 2025	Chapter 3 (Utilities and Energy Recovery)	
September 29 th 2025	Chapter 3 (Utilities and Energy Recovery)	
October 1 st 2025	Chapter 4 (Simulation)	
October 6 th 2025	Chapter 4 (Simulation)	
October 8 th 2025	Chapter 6 (Materials of construction)	
October 13 th 2025	Chapter 7 (Capital cost)	
October 15 th 2025	Chapter 7 (Capital cost)	
October 20 st 2025	Chapter 8 (Operating cost)	

October 22 nd 2025	Chapter 8 (Operating cost)	
October 27 th 2025	Chapter 9 (Economic Analysis)	
October 29 th 2025	Chapter 9 (Economic Analysis)	
November 3 rd 2025	Presentation 2	
November 5 th 2025	Presentation 2	
November 10 th 2025	Chapter 10 (Process Hazard, Professional ethics)	
November 12 th 2025	Chapter 10 (Process Hazard, Professional ethics)	
November 17 th 2025	Chapter 10 (Process Hazard, Professional ethics)	
November 19 th 2025	Professional Ethics	
November 24 th 2025	Professional Ethics	
November 26 th 2025	NO CLASS	
December 1 st 2025	Senior Design Project Presentation 1	
December 3 rd 2025	Senior Design Project Presentation 1	

Class Structure:

In a typical class,

- You are required to read the topic before the class
- During half of the class time, I will present key points of the topic
- After the presentation of the topic, I will give you a small project design that you will be working on in the class with a group of 4. In-class projects are expected to be the product of team efforts. Each member of the team is expected to contribute equally to the development of the design. In-class projects will be graded as participation. Feedback will be provided by the instructor.

Homework / mini-project assignments:

There will be four assignments in the Fall semester. For each assignment, you will be working with a group of totals of four students who will be randomly selected. Therefore, you will have the opportunity to work with different groups of people and to practice teamwork. I expect that

all the members of the group contribute equally to the group's project. Homework due dates are listed on the syllabus. They are due midnight on the due date. Late homework will be accepted up to 24 hours after the due date for 50% credit. Homework solutions will be presented in the class by students. The instructor will be randomly selecting which group will present which homework question. You should be prepared to present any questions from the homework. Also, peer evaluations will be very important. Your peers in particular homework will be evaluating your performance. Those evaluations will be graded and it will affect your homework grade. Peer evaluations will be done on catme.org and average evaluation scores of group members will be calculated. If your peer evaluation average is less than %75, your grade will be calculated by multiplying group grade by peer evaluation average percentage.

Final Project/ Proposals/Report 1 (written and oral):

A major design project will be assigned to each group of students at the end of October. Student groups will include 4 members. Each group will have its own design project. Each group is required to submit their Report 1 at the end of the semester. The Report 1 should include introduction, market analysis, competing processes, process description, design basis. Groups will present their project at the end of the semester. Spring semester mentors will be invited to class during presentations to give feedback to students on their project.

Exams/ Quizzes:

There will be NO exam for this course. NO Quizzes on presentations / lectures.

Presentations:

Students are required to give three oral presentations throughout the course. The first presentation will be based on HW 1 topic for 10 min. The presentation should be prepared for a general public audience as well as for chemical engineers. Presentations will be graded based on the design of the presentation, how you form the story, and how you tell the story. Presentations will be graded by the instructor. Classmates will be providing feedback.

Attendance /Participation :

Students are expected to attend all classes, and attendance is incorporated as 20% of the final grade. Students are allowed two excused absences and two unexcused absence. If absences exceed these limits, the student will receive a zero for attendance. Participation in class is also important. It will be incorporated as 10% of overall grade and attendance as 10% of the overall grade. In-class assignments will be graded as participation.

SaChE safety modules:

Students are expected to complete SaChE safety modules online and submit their certificates of completion to the instructor (<https://www.aiche.org/ccps/education/safety-and-chemical-engineering-education-sache-certificate-program>). There are only two required SaChE modules for ChE 396:

- Inherently safer design
- Choose one from Level III list

Grading Policy:

Course grades will be weighted according to the following grading formula.

Attendance/ Participation.....25% (in-class assignments and will be incorporated into participation)
 Homework/Mini-Project Assignments:30% (4 of them, group of 4, written; the individual grade will be calculated based on a multiplication of group grade with peer evaluation percentage, groups will be chosen randomly)
 Oral Presentations:15% (2 times, the first one is based on HW 1, group of 4, the second one is on equipment types/sizing)
 Final Project/ Proposal..... 20% (written and oral only)
 SACHE Modules /CSB reflection..... 10%
 Extra credit (professional development, company tour, Kahoot, survey)..... up to 5%

The expectation is that all students will do well in this course if they actively participate in the course, do assignments on time, and keep up with course materials.

Most relevant student outcomes for this course:

SO	Description
1	Ability to apply knowledge of mathematics, science and engineering
2	Ability to identify, formulate and solve engineering problems
3	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice
7	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
8	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
9	Knowledge of contemporary issues
10	Recognition of the need for and ability to engage in life-long learning
11	Ability to function on multidisciplinary teams
12	Ability to communicate effectively

Academic Integrity Policy

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, and administrators—share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy: <http://dos.uic.edu/community-standards/>

Academic Misconduct

Please see collaboration policies written above for each type of work for this class. In all cases, you cannot submit someone else's work (partial or complete) as your own. You are guilty of academic misconduct if you extend or receive any kind of unauthorized assistance. Absolutely no transfer of assignment between students is permitted (paper or electronic) under any circumstance, and you may not solicit code from other people, family, friends, or internet resources. Other examples of academic misconduct include emailing your work to another student, copying-pasting from the internet, working in a group on an assignment that states it is individual work, and allowing a tutor, TA, or another individual to write an answer or code for you. Academic misconduct is unacceptable. Minimum penalty will be a zero score on that assignment or assessment, with a letter drop to your final grade in the class. I reserve the right to exercise a change in the way grades are calculated if I deem it is necessary. More than one academic misconduct offense in this course will result in failure of the class. This means if you took this course a previous semester and were found guilty of academic misconduct, a second offense this semester will result in failure from the course. Similarly, if you have two offenses in the same semester, you will receive an F in course. Note that the Dean of Students will change your grade from W to an F and only with my approval can it remain as a W (if you were found guilty and drop the class). EVERY academic misconduct offense must be reported to the Dean of Students (if you ask me not to, you are asking me to not do my job, so please don't).

Religious Holidays

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure: <http://policies.uic.edu/educational-policy/student-academic-grievance-policy/>

Disability Accommodation

The University of Illinois at Chicago is committed to maintaining a barrier-free environment so that students with disabilities can fully access programs, courses, services, and activities at UIC. Students with disabilities who require accommodations for access to and/or participation in this course are welcome, but must be registered with the Disability Resource Center (DRC). You

may contact DRC at 312-413-2183 (v) or 773-649-4535 (VP/Relay) and consult the following: <http://drc.uic.edu/guide-to-accommodations>. If you have a letter of accommodation, it is best to stop by my office hours to talk about it in person. Any exams scheduled in the DRC must overlap with the originally scheduled exam. In other words, you must start the exam before or at the same time as the originally scheduled exam. Additionally, you may not leave the DRC exam room until after the originally scheduled exam is over or later.

Mental Health and Emotional Wellness

We value your mental health and emotional wellness as part of the UIC student experience. The UIC Counseling Center offers an array of services to provide additional support throughout your time at UIC, including workshops, peer support groups, counseling, self-help tools, and initial consultations to speak to a mental health counselor about your concerns. Please visit the Counseling Center website for more information (<https://counseling.uic.edu/>). Further, if you think emotional concerns may be impacting your academic success, please contact your faculty and academic advisers to create a plan to stay on track.

Campus Advocacy Network

If you have experienced sexual assault, domestic/dating violence, family abuse, stalking, or harassment, there is help available on campus. The Campus Advocacy Network provides free and confidential services to UIC students, faculty, and staff who have experienced or are experiencing interpersonal violence. If you would like to connect with an advocate, please contact CAN at can-appointment@uic.edu or call (312) 413-8206. To learn more, visit CAN's website at <https://can.uic.edu/>.